SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO2 ASSESSMENT TOOL 1 – Scenario-Based Requirement Analysis

Course Code:	CSA10 / CSA1005	**Course Title:**	Software Engineering
CO Assessed:	CO2	**Assessment:**	Assessment Tool 2 – Scenario-Based Requirement Analysis
Weightage:	20%	**Total Marks:**	25
CO2 Statement:	*Perform detailed requirements analysis, design scalable architectures, and prioritize features with a focus on cross-disciplinary skills*		
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Date:	27.10.26	**Duration:**	60 minutes

Code of Conduct

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

Annexure A – Scenario-Based Requirement Analysis

Instructions to Students:

Each student will be assigned an individual software project problem statement. Analyze the given scenario and prepare a complete Software Requirement Analysis document by following the steps below.

Question

Based on the **assigned problem statement**, perform a **Scenario-Based Requirement Analysis** and prepare a Software Requirement Specification (SRS) document. Your analysis should include the following:

1. **Study and Analyze the Scenario**
   - Carefully understand the given problem statement.
   - Identify the business domain, stakeholders, existing challenges, and system objectives.
2. **Identify Functional and Non-Functional Requirements**
   - List all functional requirements required by the proposed system.
   - Identify suitable non-functional requirements such as performance, security, reliability, usability, scalability, maintainability, and availability.
3. **Identify Actors and Use Cases**
   - Identify all primary and secondary actors interacting with the system.
   - List the major use cases associated with each actor.
   - Draw a Use Case Diagram representing the interactions between actors and the system.
4. **Prepare the Software Requirement Specification (SRS)**
   Develop an SRS document containing:
   - Introduction
   - Problem Description
   - Scope of the System

- Functional Requirements
- Non-Functional Requirements
- Use Case Descriptions
- Data Requirements
- External Interface Requirements
- System Features

5. **Identify Assumptions and Constraints**
   - List the assumptions considered while developing the system.
   - Identify technical, operational, economic, legal, and time-related constraints affecting the project.
6. **Validate Requirement Completeness and Consistency**
   - Verify whether all stakeholder requirements have been addressed.
   - Check for ambiguity, redundancy, inconsistency, and missing requirements.
   - Suggest improvements to enhance the quality and completeness of the requirement specification.

Rubrics for Calculation

Criteria	Excellent (4 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Scenario Understanding and Problem Analysis (30%)	Demonstrates thorough understanding of the scenario, stakeholders, objectives, and business context with clear analysis.	Demonstrates good understanding with minor omissions.	Basic understanding with limited analysis.	Poor understanding; major aspects missing or incorrect.
Functional and Non-Functional Requirement Identification (25%)	Clearly identifies comprehensive, accurate, and well-classified functional and non-functional requirements.	Identifies most requirements with minor classification errors.	Identifies only basic requirements; several omissions.	Requirements are incomplete, inaccurate, or poorly classified.
Actor Identification and Use Case Modelling (20%)	Correctly identifies all relevant actors and use cases; use case diagram is complete, accurate, and follows UML standards.	Most actors and use cases identified; diagram has minor errors.	Limited actors/use cases identified; diagram partially correct.	Actors/use cases missing or incorrect; diagram absent or inaccurate.
Software Requirement Specification (SRS) Documentation	SRS is well-structured, complete, professionally organized, and includes all required	SRS is organized with minor missing sections or formatting issues.	SRS includes only essential sections with limited detail.	SRS is incomplete, poorly organized, or lacks essential

(15%)	sections.			components.
Assumptions, Constraints, and Requirement Validation (10%)	Clearly identifies realistic assumptions and constraints; validates requirements for completeness, consistency, and ambiguity with appropriate justification.	Identifies most assumptions and constraints; validation is adequate.	Limited assumptions/constraints; validation is superficial.	Assumptions, constraints, and validation are missing or incorrect.

Criteria	**Mark**
Scenario Understanding and Problem Analysis (30%)	/5
Functional and Non-Functional Requirement Identification (25%)	/5
Actor Identification and Use Case Modelling (20%)	/5
Software Requirement Specification (SRS) Documentation (15%)	/5
Assumptions, Constraints, and Requirement Validation (10%)	/5
TOTAL	/25

Smart Campus Lost-and-Found Management System.

1. Scenario Understanding and Problem Analysis:

1.1 Business Domain

The project belongs to the campus administration and student-support domain. It provides a centralized digital platform for reporting, searching, matching, and recovering personal belongings lost or found within a college campus.

1.2 Problem Description

Students and staff frequently lose belongings such as ID cards, wallets, electronic devices, books, and accessories. Conventional processes rely on notice boards, word-of-mouth, and manually maintained records. These methods make reports difficult to search, create duplicate or incomplete records, delay communication, and provide limited evidence of ownership before an item is returned.

1.3 Stakeholders

- Students and staff: report lost items, register found items, search records, and recover belongings.
- Campus administrator or lost-and-found coordinator: moderate reports, verify users or claims, resolve disputes, and monitor records.
- Notification service: delivers status updates and possible-match alerts.
- Campus management: benefits from reports and analytics about recovery rates and common loss locations.

1.4 Objectives

- Allow users to register lost and found items with descriptions, photographs, locations, and timestamps.
- Enable searching and comparison of lost and found reports to identify potential matches.
- Notify relevant users when a possible match or status change is identified.
- Support ownership verification before handover.
- Maintain recovery history and generate basic lost-and-found analytics.

1.5 Existing Challenges

- Scattered notices and manual records are hard to maintain and search.
- Descriptions may be incomplete, making matching difficult.
- Communication between the person who lost an item and the person who found it can be delayed.
- Ownership may not be checked consistently before handover.
- Manual records provide limited recovery history and reporting.

2. Scope of the System

The system will provide a web-based or mobile-friendly interface for authorized campus users. It will support account access, lost/found reporting, item search, possible-match identification, notifications, claim verification, recovery records, and administrator reporting.

Out of scope: Police investigations, automatic proof of legal ownership, physical storage of items, and guaranteed identification of a match. The system suggests possible matches; users or an administrator must verify claims before handover.

3. Functional Requirements

ID	Requirement	Description
FR-01	User access	The system shall allow campus users to register, log in, and log out.
FR-02	Lost-item report	The system shall allow a user to submit an item description, category, date/time lost, location, and optional photograph.
FR-03	Found-item report	The system shall allow a user to submit a found-item description, category, date/time found, location, and optional photograph.
FR-04	Search and filter	The system shall allow users to search and filter reports by keyword, category, location, and date.
FR-05	Potential matching	The system shall compare lost and found reports using available attributes such as category, description, location, and date, and display possible matches.
FR-06	Notifications	The system shall notify relevant users when a possible match or report status update is available.
FR-07	Claim and verification	The system shall allow a claimant to provide identifying details and shall support administrator or authorized coordinator verification before handover.
FR-08	Status management	The system shall allow authorized users to update report status, for example Open, Potential Match, Under Verification, Recovered, or Closed.
FR-09	Recovery history	The system shall retain records of recovered items, handover status, and relevant timestamps.
FR-10	Administration	The administrator shall be able to review, correct, hide, or close inappropriate, duplicate, or resolved reports.
FR-11	Analytics	The system shall summarize reports, recoveries, unresolved items, and common reported locations.
FR-12	Privacy controls	The system shall restrict sensitive claimant details and contact information to authorized users.

4. Non-Functional Requirements

Quality Attribute	Requirement
Performance	Common pages and searches should respond within 3 seconds under normal campus usage.
Security	Passwords must be stored securely using a suitable password-hashing method; access shall be role-based and protected by authenticated sessions.

Privacy	Personal contact details and sensitive item information shall not be publicly exposed unnecessarily.
Usability	Forms and search screens shall use clear labels, validation messages, and simple navigation.
Reliability	The system shall validate required fields and avoid losing successfully submitted reports during normal operation.
Availability	The system should be available whenever campus users need to report or search for items, except during planned maintenance.
Scalability	The design should support an increase in users and reports without requiring a complete redesign.
Maintainability	The application shall use modular code, documented requirements, and consistent data structures.
Compatibility	The interface should work on current common desktop and mobile browsers.
Data integrity	The system shall preserve report status and timestamps and prevent unauthorized alteration of records.

5. Actors and Use Cases :

5.1 Actors

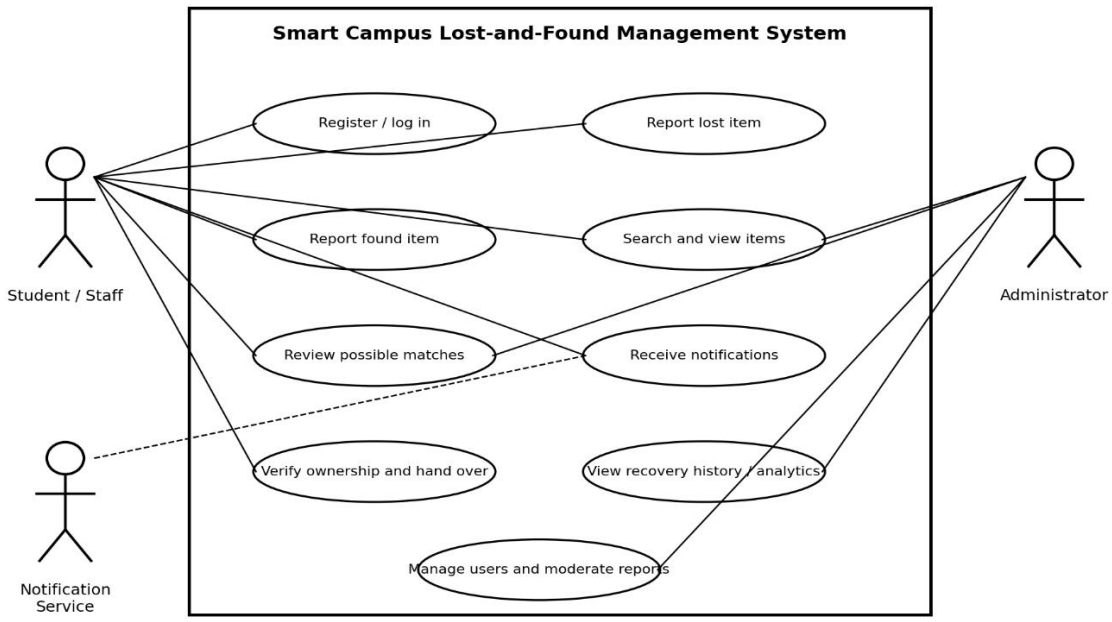
Actor	Type	Responsibilities
Student / Staff	Primary	Register or log in, report lost/found items, search reports, submit claims, view alerts, and confirm recovery.
Administrator / Coordinator	Primary	Moderate reports, verify claims, manage report status, resolve duplicates, and view analytics.
Notification Service	Supporting / external	Send match alerts and report-status notifications to users.

5.2 Major Use Cases by Actor

Actor	Major Use Cases
Student / Staff	Register / log in; report lost item; report found item; search and view items; review possible matches; receive notifications; submit ownership claim; confirm recovery.
Administrator / Coordinator	Review and moderate reports; verify claimant and ownership evidence; update report status; resolve duplicates or disputes; view recovery history and analytics.
Notification Service	Deliver possible-match alerts and status notifications.

5.3 Use Case Diagram

The diagram shows the main interactions. Student/staff users create and search reports and participate in recovery; the administrator moderates records and verifies claims; the notification service supports alert delivery.



6. Use Case Descriptions:

UC-01: Report a lost item

- Primary actor: Student / Staff
- Main flow: 1. User logs in. 2. Opens the lost-item form. 3. Enters description, category, location, date/time and optional photo. 4. System validates and saves the report. 5. System confirms submission and checks for possible matches.
- Postcondition: Report is saved with a unique ID and Open status.
- Alternative / exception: Required fields missing or invalid: system shows an error and requests correction.

UC-02: Report a found item

- Primary actor: Student / Staff
- Main flow: 1. User logs in. 2. Opens the found-item form. 3. Enters item details, found location/date/time and optional photo. 4. System validates and saves the report. 5. System compares it with lost reports.
- Postcondition: Found report is stored and potential matches can be reviewed.
- Alternative / exception: Invalid or incomplete details: user is asked to correct the form.

UC-03: Search and review possible matches

- Primary actor: Student / Staff

- Main flow: 1. User enters keywords or filters. 2. System displays matching reports. 3. User opens a report and reviews non-sensitive details. 4. User follows the claim/contact procedure if an item may match.
- Postcondition: Relevant reports and possible matches are displayed.
- Alternative / exception: No results: system displays a clear no-results message and allows revised search.

UC-04: Verify claim and hand over item

- Primary actor: Administrator / Coordinator
- Main flow: 1. Coordinator reviews the claim. 2. Claimant provides distinguishing details or evidence not fully disclosed publicly. 3. Coordinator checks the evidence and report. 4. If acceptable, coordinator authorizes handover and updates status. 5. System records recovery.
- Postcondition: Verified handover is recorded and the report is marked Recovered or Closed.
- Alternative / exception: Evidence is insufficient or conflicting: do not authorize handover; request more evidence or escalate to the coordinator.

UC-05: Receive a notification

- Primary actor: Notification Service
- Main flow: 1. System detects a possible match or status update. 2. System prepares a notification. 3. Service delivers it using the configured channel. 4. Delivery outcome is recorded where supported.
- Postcondition: Relevant user is notified of the possible match or update.
- Alternative / exception: Delivery failure: retain the event for retry or show the update in the user account.

7. Data Requirements:

Data Entity	Key Data Fields	Purpose
User	User ID, name, campus email/ID, role, authentication data, account status	Identify users and apply role-based access.
Item Report	Report ID, report type (lost/found), title, description, category, date/time, location, image reference, reporter ID, status, created timestamp	Store lost and found reports.
Match Record	Match ID, lost report ID, found report ID, matching attributes/score, review status, created timestamp	Track possible matches and their review.
Claim / Verification	Claim ID, report ID, claimant ID, evidence notes, reviewer ID, decision, decision timestamp	Support safe claim verification and auditability.
Notification	Notification ID, recipient ID, event type, message reference, created time, delivery status	Track match and status alerts.
Recovery Record	Recovery ID, report ID, verified by, handover timestamp, final status	Maintain recovery history and analytics.

8. External Interface Requirements

Interface	Requirement
User interface	Responsive screens for login, lost/found report forms, search results, report details, claim submission, notifications, and user history.
Administrator interface	Restricted dashboard for report moderation, claim review, status changes, and analytics.
Notification interface	Integration with a configured campus email or in-app notification service; exact provider depends on deployment.
Image upload interface	Accept supported image formats and enforce file-size/type validation before storage.
Database interface	Application communicates with a database through validated queries or a suitable data-access layer; direct public access to the database is not allowed.
Browser/device interface	Support current mainstream desktop and mobile browsers with readable layouts and accessible form controls.

9. System Features and Priority

Feature	Priority	Reason
Account access and role control	Must have	Protects reports, claimant details, and administrative actions.
Lost/found report submission	Must have	Core function needed to create searchable records.
Search and filtering	Must have	Allows users to find relevant reports.
Possible-match identification	Must have	Reduces manual effort in connecting lost and found items.
Claim verification and recovery status	Must have	Helps prevent incorrect handover.
Notifications	Should have	Reduces delay in informing users about matches and status changes.
Recovery history and analytics	Should have	Supports follow-up and campus-level reporting.
Advanced image recognition or mobile push alerts	Could have	Potential future enhancement; not required for the initial version.

10. Assumptions and Constraints:

10.1 Assumptions

- Users have access to a campus email or another approved method of account verification.

- Users provide truthful descriptions and accurate locations and timestamps to the best of their knowledge.
- A campus coordinator or authorized administrator is available to review disputed claims and approve handovers.
- Users can upload a photograph, but a photograph is optional because some items may not have one.
- The institution provides an agreed policy for data retention, privacy, and disposal of unclaimed items.

10.2 Constraints

Constraint Type	Constraint
Technical	Implementation depends on available hosting, database, image storage, and notification services; match results depend on the quality of entered data.
Operational	The system cannot physically store items; the campus must define where found items are kept and who handles handover.
Economic	The initial version should use available or low-cost tools and avoid unnecessary paid services.
Legal / privacy	Personal data, photographs, and claimant evidence must be handled according to applicable privacy rules and institutional policy.
Time-related	The assessment prototype must prioritize core reporting, searching, matching, verification, and status tracking before optional enhancements.

11. Requirement Completeness and Consistency Validation

Validation Check	Result / Action
Stakeholder coverage	Students/staff, administrator/coordinator, notification service, and campus management needs are represented.
Functional coverage	Reporting, searching, matching, notifications, verification, status management, recovery history, administration, and analytics are specified.
Non-functional coverage	Performance, security, privacy, usability, reliability, availability, scalability, maintainability, compatibility, and data integrity are addressed.
Ambiguity	“Potential match” is treated as a suggestion, not a confirmed identity. A human verification step is required before handover.
Consistency	Only authorized users/admins can change sensitive statuses; reports use defined statuses and unique identifiers.

Redundancy	Lost and found reports share common fields while retaining a report-type field, avoiding separate inconsistent data definitions.
Missing requirements to confirm	Before deployment, confirm notification channel, data-retention period, supported file limits, administrator approval workflow, and procedure for unclaimed items.
Suggested improvements	Define acceptance tests, access-control matrix, backup/recovery policy, and measurable service targets before production deployment.

12. Conclusion

The proposed Smart Campus Lost-and-Found Management System replaces scattered manual notices with a centralized process for reporting, searching, identifying possible matches, notifying users, verifying claims, and recording recovery. The requirements distinguish automated suggestions from verified ownership and include security, privacy, operational assumptions, and constraints. The specification can be validated further with campus stakeholders before implementation.

13. Suggested Acceptance Criteria

- A valid lost or found report is saved and assigned a unique report ID.
- A user can search reports using keywords and at least category, location, and date filters.
- When reports share relevant attributes, the system displays them as possible matches without declaring ownership confirmed.
- A claim cannot be marked verified or a handover completed without an authorized verification action.
- Users cannot access restricted administrator functions or another user’s sensitive claim details.
- Recovery history and summary counts reflect status changes accurately.